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Method and system for manufacturing a semiconductor device

Abstract

A method for manufacturing a semiconductor device includes forming a photoresist layer that includes a photoresist composition over a wafer to produce a photoresist-coated wafer. The photoresist layer is selectively exposed to actinic radiation to form a latent pattern in the photoresist layer. The latent pattern is developed by applying a developer to the selectively exposed photoresist layer under a first pressure gas flow setting in a development chamber. The photoresist layer is rinsed, under the first pressure gas flow setting, to form a patterned photoresist layer exposing a portion of the wafer in the development chamber. The patterned photoresist layer is spin dried under a second pressure gas flow setting. A pressure of the development chamber under the second pressure gas flow setting is greater than the pressure of the development chamber under the first pressure gas flow setting.

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Background/Summary

BACKGROUND

(1) During an integrated circuit (IC) design, a number of patterns of the IC, for different steps of IC processing, are generated on a substrate. The patterns may be produced by projecting, e.g., imaging, layout patterns of a mask on a photo resist layer of the wafer. A lithographic process transfers the layout patterns of the masks to the photo resist layer of the wafer such that etching, implantation, or other steps are applied only to predefined regions of the wafer. It is desirable that the layout patterns are produced on the substrate with no errors such that etching produces no defects on the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. **1**A and **1**B show a process flow for generating an etched pattern on a semiconductor substrate and a process flow for development.

(3) FIG. **2** shows a process stage of a sequential operation according to an embodiment of the disclosure.

(4) FIG. **3** shows a schematic view of an exposure device of a lithography system for generating a photo resist pattern on a wafer.

(5) FIGS. **4**A and **4**B show a process stage of a sequential operation according to an embodiment of the disclosure.

(6) FIG. **5** shows a process stage of a sequential operation according to an embodiment of the disclosure.

(7) FIGS. **6**A and **6**B show a process stage of a sequential operation according to an embodiment of the disclosure.

(8) FIGS. **7**A, **7**B, **7**C, **7**D, and **7**E show a system for generating a resist pattern and/or an etched pattern on the substrate, developed material residues and water residues on the substrate, a defect caused by the residues in an etched pattern on the substrate, and a graph of the number of pattern collapses on a wafer vs. the spin drying speed.

(9) FIGS. **8**A, **8**B, **8**C, **8**D, **8**E, **8**F, and **8**G show a system for generating a resist pattern and/or an etched pattern on the substrate, an exhaust control system of the system for generating the resist pattern and/or the etched pattern on the substrate, and developed material residues and water residues with and without using the exhaust control system in accordance with some embodiments of the present disclosure.

(10) FIGS. **9**A and **9**B show a process stage of a sequential operation according to an embodiment of the disclosure.

(11) FIGS. **10**A and **10**B show an inspection system of the residue or defects on a surface of the substrate and controlling the system for generating a resist pattern and/or an etched pattern on the substrate in accordance with some embodiments of the present disclosure.

(12) FIG. **11** shows a control system for generating a resist pattern and/or an etched pattern on the substrate in accordance with some embodiments of the present disclosure.

(13) FIG. **12** shows a flow diagram of a process for generating a resist pattern and/or an etched pattern on a substrate in accordance with some embodiments of the present disclosure.

(14) FIGS. **13**A and **13**B illustrate an apparatus for controlling the system for generating a resist pattern and/or an etched pattern on the substrate in accordance with some embodiments of the present disclosure.

(15) FIG. **14** shows a process stage of a sequential operation according to an embodiment of the disclosure.

(16) FIGS. **15**A and **15**B show a process stage of a sequential operation according to an embodiment of the disclosure.

(17) FIG. **16** shows a process stage of a sequential operation according to an embodiment of the disclosure.

(18) FIGS. **17**A and **17**B show a process stage of a sequential operation according to an embodiment of the disclosure.

(19) FIGS. **18**A and **18**B show a process stage of a sequential operation according to an embodiment of the disclosure.

DETAILED DESCRIPTION

(20) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(21) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In addition, the term “being made of” may mean either “comprising” or “consisting of.” In the present disclosure, a phrase “one of A, B and C” means “A, B and/or C” (A, B, C, A and B, A and C, B and C, or A, B and C), and does not mean one element from A, one element from B and one element from C, unless otherwise described.

(22) Remaining water residues and developed material residues on the developed photo resist pattern on the surface of the substrate, e.g., a wafer, may interfere with processes that occur after the photo resist development process. The remaining water residues and developed material residues may interfere with subsequent etching processes and cause a number of pinching and bridging in the connection lines or other patterns, e.g., a line collapse or a pattern collapse, in the developed resist pattern that is produced on the substrate. The remaining water residues and developed material residues on the developed resist pattern on the surface of the substrate can be reduced by spinning the substrate, in a spin drying operation at the end of the development process, to use the centrifuge force to move the remaining water residues and developed material residues farther from the center of the substrate to the edge of the substrate and cause the remaining water residues and developed material residues pushed off the edge of the substrate. If the spinning is not enough to remove all the remaining water residues and developed material residues from the surface and/or edge of the substrate, the remaining water residues and developed material residues can further be reduced by providing a gas flow, e.g., an air flow, that passes over the substrate during the spin drying operation. In some embodiments, by increasing the gas flow (e.g., by increasing the gas flow pressure) the remaining water residues and developed material residues are pushed further to the edge of the substrate and off the edge of the substrate such that and the water residues and developed material residues are essentially eliminated from the surface, e.g., top surface, of the substrate.

(23) FIGS. **1**A and **1**B show a process **100** for generating an etched pattern on a semiconductor substrate and a development operation S**112**, e.g., developing a resist pattern on the semiconductor substrate. In some embodiments, the process **100** is performed by a lithography system that is controlled by a control and/or computer system, e.g., a control system **700** of FIG. **11** and/or a computer system **900** of FIGS. **13**A and **13**B. In a wafer loading operation S**101**, a substrate, e.g., a semiconductor wafer, is loaded into a semiconductor device processing tool. In some embodiments, the tool is a coater-developer tool (not shown). In some embodiments in a resist coating operation S**102**, a photoresist layer **15** of a resist material is disposed, e.g., coated, on a top surface of a semiconductor substrate **10** shown in FIG. **2**, e.g., the wafer or a work piece, to produce a

photoresist-coated wafer. As shown in FIG. **2**, the photoresist layer **15** is disposed over a semiconductor substrate **10**, to provide a photoresist coated wafer **162**. In some embodiments, the photoresist is applied using a process such as a spin-on coating process, a dip coating method, an air-knife coating method, a curtain coating method, a wire-bar coating method, a gravure coating method, a lamination method, an extrusion coating method, combinations of these, or the like. In some embodiments, a thickness of the photoresist layer **15** ranges from about 10 nm to about 300 nm. At photoresist thicknesses below the disclosed range there may be insufficient photoresist coverage to protect the underlying substrate during subsequent etching operations. At photoresist thicknesses greater than the disclosed range, there may be excessive photoresist waste and longer processing time.

(24) A pre-exposure (or post application bake (PAB)) is performed at a PAB operation S**104** and the semiconductor substrate **10** including the photoresist layer **15** is baked to drive out solvent in the resist material and solidify the photoresist layer **15** on top of the semiconductor substrate **10**. In some embodiments, the photoresist layer **15** is heated at a temperature of about 40° C. to about 120° C. for about 10 seconds to about 10 minutes during PAB operation S**104**. In the present disclosure, the terms resist and photoresist are used interchangeably. Selecting a PAB time and/or temperature below the above range may be insufficient for the removal of photoresist solvent the resist material. Selecting a PAB time and/or temperature greater than the above range, may increase energy costs and longer processing times.

(25) In an exposure operation S**108**, the photoresist-coated substrate is loaded into a photolithography tool **200**, as shown in FIG. **3**. The photoresist layer **15** is exposed to actinic radiation, in the photolithography tool **200**. The exposure operation S**108** also projects a layout pattern of a mask, e.g. a photomask **205***c* of FIG. **3**, using the actinic radiation from the radiation source, onto the photoresist layer **15** on the semiconductor substrate **10**. In some embodiments, the photoresist layer **15** is selectively exposed to ultraviolet radiation. In some embodiments, the ultraviolet radiation is deep ultraviolet radiation (DUV). In some embodiments, the ultraviolet radiation is extreme ultraviolet (EUV) radiation. In some embodiments, the radiation is an electron beam. In the present disclosure, the terms mask, photomask, and reticle are used interchangeably.

(26) In some embodiments, the photomask **205***c* is a reflective mask and the layout pattern on the mask is projected by extreme ultraviolet (EUV) radiation from an EUV light source **109** onto the photoresist layer **15** to generate a latent image in the photoresist layer **15** on the semiconductor substrate **10**.

(27) A post exposure bake (PEB) is performed at a PEB operation S**110** on the substrate where the resist layer is further baked after being exposed to the actinic radiation and before being developed in the development operation S**112**. In some embodiments, the photoresist layer **15** is heated to a temperature of about 50° C. to about 160° C. for about 20 seconds to about 10 minutes. In some embodiments, the photoresist layer **15** is heated for about 30 seconds to about 5 minutes. In some embodiments, the photoresist layer **15** is heated for about 1 minute to about 2 minutes. The post-exposure baking may be used to assist in the generating, dispersing, and reacting of acid/base/free radical generated from the impingement of the actinic radiation upon the photoresist layer **15** during the exposure. Such assistance helps to create or enhance chemical reactions that generate chemical differences between the exposed region and the unexposed region within the photoresist layer. These chemical differences also cause differences in the solubility between the exposed region and the unexposed region. At PEB times and temperatures below the disclosed ranges there may be insufficient generation, dispersion, and reaction of the chemical reactants in the exposed region of the photoresist layer. At PEB times and temperatures greater than the disclosed ranges, there may be increased energy costs and longer processing times, and thermal degradation of the photoresist layer.

(28) The selectively exposed photoresist layer is subsequently developed by applying a developer to the selectively exposed photoresist layer in operation S**112**. As shown in FIG. **5**, a development

material **57**, e.g., a developer, is supplied from a nozzle **62** to the photoresist layer **15**. In some embodiments, the exposed region **50** of the photoresist layer radiation is removed by the development material **57** forming a pattern of openings **55***a* in the photoresist layer **15** to expose the semiconductor substrate **10**, as shown in FIG. **6**A to produce the patterned semiconductor substrate **163**. In other embodiments, the unexposed region **52** of the photoresist layer is removed by the development material **57** forming a pattern of openings **55***b* in the photoresist layer **15** to expose the semiconductor substrate **10**, as shown in FIG. **6**B to produce the patterned semiconductor substrate **165**.

(29) In some embodiments, the development material **57** includes a solvent, and an acid or a base. In some embodiments, the concentration of the solvent is from about 60 wt. % to about 99 wt. % based on the total weight of the photoresist developer. The acid or base concentration is from about 0.001 wt. % to about 20 wt. % based on the total weight of the photoresist developer. In certain embodiments, the acid or base concentration in the developer is from about 0.01 wt. % to about 15 wt. % based on the total weight of the photoresist developer. At chemical component concentrations below the disclosed range there may be insufficient development. At chemical component concentrations greater than the disclosed range, there may be overdevelopment and waste of materials.

(30) In some embodiments, the development material **57** is applied to the photoresist layer **15** using a spin-on coating process. In the spin-on coating process, the development material **57** is applied to the photoresist layer **15** from above the photoresist layer **15** while the photoresist-coated substrate is rotated, as shown in FIG. **5**. In some embodiments, the development material **57** is supplied at a rate of between about 5 ml/min and about 800 ml/min, while the photoresist coated semiconductor substrate **10** is rotated at a speed of between about 100 rpm and about 2000 rpm. In some embodiments, the developer is at a temperature of between about 10° C. and about 80° C. The development operation continues for between about 30 seconds to about 10 minutes in some embodiments. At spin rates, times, and temperatures below the disclosed range there may be insufficient development. At spin rates, times, and temperatures greater than the disclosed range, there may be degradation of the resist pattern.

(31) While the spin-on coating process is one suitable method for developing the photoresist layer **15** after exposure, it is intended to be illustrative and is not intended to limit the embodiment. Rather, any suitable development operations, including dip processes, puddle processes, and spray-on methods, may alternatively be used. All such development operations are included within the scope of the embodiments.

(32) FIG. **1**B shows the operation S**112** for development. In some embodiments, the operation S**112** is performed by a resist development system, e.g., a development system **800** of FIG. **8**A, that is controlled by the control system **700** of FIG. **11** and/or the computer system **900** of FIGS. **13**A and **13**B. In a pre-wet operation S**152**, the photoresist-coated wafer **162** loaded on a stage is rinsed such that the photoresist layer **15** on the semiconductor substrate **10** that is loaded on a stage of a development system is rinsed with water, e.g., de-ionized (DI) water. A development system **800** is described with respect to FIG. **8**A. In an apply developer solution operation S**154**, as shown in FIG. **5**, a development material **57** is applied, via a nozzle **62**, to the photoresist material on the surface of the semiconductor substrate **10**. The development material **57** is applied when the semiconductor substrate **10** is on a stage **825**, e.g., a rotatable wafer stage, of the development system **800**. In a water rinse operation S**156**, the developed material e.g., the photoresist developed material, on the surface of the semiconductor substrate **10** is rinsed with water, e.g., DI water to produce a resist pattern on the surface of the semiconductor substrate **10**. In some embodiment, after the water rinse operation S**156** some water residues and developed material residues remain on the surface of the semiconductor substrate **10**. In a spin dry operation S**158**, the substrate is rotated on the stage such that the centrifuge force moves the remaining water residues and developed material residues on the semiconductor substrate **10** farther from the center of the semiconductor substrate **10** to the

edge of the semiconductor substrate **10** and cause part of the remaining water residues and developed material residues to be pushed off from the edge of the substrate and, thus, reducing the remaining water residues and developed material residues on the surface of the semiconductor substrate **10**. After the spin dry operation S**158**, a wafer **164** with the developed resist pattern is sent for the next process. In some embodiments, during the spin dry operation S**158**, the substrate is rotated at a speed between about 100 rpm and about 3000 rpm.

(33) FIG. **3** shows a schematic view of a photolithography tool **200** for generating a resist pattern on a wafer. In some embodiments, the photolithography tool **200** is an extreme ultraviolet (EUV) lithography tool, where the photoresist layer **15** is exposed by a patterned beam **31** of EUV radiation. A chamber of the photolithography tool **200** may include a wafer movement device, e.g., a stage **160**, a stepper, a scanner, a step and scan system, a direct write system, a device using a contact and/or proximity mask, etc. The tool is provided with one or more optics **205***a*, **205***b*, for example, to illuminate a patterning optics, such as a reticle, e.g., a reflective mask **205***c* with a radiation beam **29**, e.g., an EUV radiation beam in some embodiments. The illumination of the patterning optics may produce a patterned beam **31**. One or more reduction projection optics **205***d*, **205***e*, of the optical system projects the patterned beam **31** onto a photoresist layer **15** of the semiconductor substrate **10**. A stage controller **170** may be coupled to the wafer movement device, e.g., the stage **160**, for generating a controlled relative movement between the semiconductor substrate **10** and the patterning optics, e.g., the reflective mask **205***c*. By the controlled relative movement, different dice of the semiconductor substrate **10** are patterned. In some embodiments, the reflective mask **205***c* is mounted on a reticle stage **401**, e.g., a mask stage.

(34) As further shown, the photolithography tool **200** of FIG. **3** includes a radiation source **109** to generate the radiation beam **29** used to irradiate the reflective mask **205***c*. Because gas molecules absorb EUV light, the photolithography tool **200** is maintained under a vacuum environment to avoid EUV intensity loss. In addition, in some embodiments, the photolithography tool **200** includes an exposure controller **430** to control an intensity of the radiation beam **29**. In some embodiments, the exposure controller **430** adjusts the intensity of the radiation by adjusting a projection time of the lithography operation to pattern the resist layer. In some embodiments, a pressure inside the photolithography tool **200** is sensed by a pressure sensor **408** inside the photolithography tool **200** and is controlled by a vacuum pressure controller **406** that is coupled to the photolithography tool **200**.

(35) As shown in FIG. **4**A, the exposure radiation beam **29** passes through a photomask **30** before irradiating the photoresist layer **15** in some embodiments. In some embodiments, the photomask has a pattern to be replicated in the photoresist layer **15**. The pattern is formed by an opaque pattern **35** on the photomask substrate **40**, in some embodiments. The opaque pattern **35** may be formed by a material opaque to ultraviolet radiation, such as chromium, while the photomask substrate **40** is formed of a material that is transparent to ultraviolet radiation, such as fused quartz.

(36) In some embodiments, where the exposure radiation is EUV radiation, the reflective photomask **205***c* is used to form the patterned exposure light, as shown in FIG. **4**B. In some embodiments, the exposure is performed in the photolithography tool **200**. The reflective photomask **205***c* includes a low thermal expansion glass substrate **70**, on which a reflective multilayer **75** of Si and Mo is formed. A capping layer **80** and absorber layer **85** are formed on the reflective multilayer **75**. A rear conductive layer **90** is formed on the back side of the low thermal expansion glass substrate **70**. In extreme ultraviolet lithography, extreme ultraviolet radiation beam **29** is directed towards the reflective photomask **205***c* at an incident angle of about 6°. A portion **31** of the EUV radiation is reflected by the Si/Mo multilayer **75** towards the photoresist-coated semiconductor substrate **10**, while the portion of the extreme ultraviolet radiation incident upon the absorber layer **85** is absorbed by the photomask. In some embodiments, additional optics, including mirrors, are between the reflective photomask **205***c* and the photoresist-coated substrate.

(37) The exposed region **50** of the photoresist layer to radiation undergoes a chemical reaction

thereby changing its solubility in a subsequently applied developer relative to the unexposed region **52** of the photoresist layer to radiation. In some embodiments, the exposed region **50** of the photoresist layer to radiation undergoes a reaction making the exposed portion more soluble in a developer. In other embodiments, the exposed region **50** of the photoresist layer to radiation undergoes a crosslinking reaction making the exposed portion less soluble in a developer.

(38) In some embodiments, the actinic radiation beam **29**, includes g-line (wavelength of about 436 nm), i-line (wavelength of about 365 nm), ultraviolet radiation, far ultraviolet radiation, extreme ultraviolet, electron beam, or the like. In some embodiments, the radiation source **109** is selected from the group consisting of a mercury vapor lamp, xenon lamp, carbon arc lamp, a KrF excimer laser light (wavelength of 248 nm), an ArF excimer laser light (wavelength of 193 nm), an F.sub.2 excimer laser light (wavelength of 157 nm), or a CO.sub.2 laser-excited Sn plasma (extreme ultraviolet, wavelength of 13.5 nm). In some embodiments, the exposure of the photoresist layer **15** uses an immersion lithography technique. In such a technique, an immersion medium is placed between the final optics and the photoresist layer, and the exposure radiation passes through the immersion medium.

(39) FIGS. **7**A, **7**B, **7**C, **7**D, and **7**E show a system for generating a resist pattern and/or an etched pattern on the substrate, developed material residues and water residues on the substrate, a defect caused by the residues in an etched pattern on the substrate, and a graph of the number of pattern collapses on a wafer vs. the spin drying speed. FIG. **7**A shows a system **300** for performing the apply developer solution operation S**154** and the water rinsing operation S**156** of the substrate **220**, where the substrate **220** is consistent with the semiconductor substrate **163** or **165**. As described with respect to FIG. **5**, before the water rinsing operation S**156**, a development material, consistent with the development material **57**, is applied in an apply developer solution operation S**154** to the surface of the substrate **220** through a nozzle (not shown in FIG. **7**A), consistent with the nozzle **62** in FIG. **5**. In the water rinsing operation S**156** of the substrate **220**, the developed material **212** that may include deionized (DI) water is moved in a direction **214** to the edges of the substrate **220** by centrifugal force caused by rotation **255** of the substrate **220**. In some embodiments, as shown in FIG. **7**B, the centrifugal force is not strong enough to remove all of the developed material from the top surface of the substrate **220** and developed material residues **224** that may include DI water remain on the top surface of the substrate **220**. In some embodiments, the developed material residues **224** are clustered in a central region **222** of the top surface of the substrate **220** but an edge region **226** of the top surface of the substrate **220** is essentially free of the development material. In some embodiments, as shown in FIG. **7**A, the back side of the substrate **220** is cleaned by DI water **210** that is delivered through the nozzles **208** tilted at an acute angle relative to the back side surface of the substrate **220**. In some embodiments, cleaning the back side of the substrate **220** causes the DI water **210** to splash, e.g., climb, over the edges of the substrate **220** and even get to the edge region **226** of the substrate **220**. The climbed over DI water may generate a barrier force in a direction **216** that blocks the developed material **206** from leaving the edge region **226** and generate developed material residues **228** that may include DI water that splashes to the edge region **226** of the top surface of the substrate **220**. FIG. **7**D shows defects **232** and **234** generated after a subsequent etching operation in an etched pattern **230** because some of the developed material **206** or **212** remaining on the top surface of the substrate **220**. As described above, the centrifugal force caused by the rotation **255** of the substrate **220** may remove the developed material **206** or **212** from the top surface of the substrate **220**. In some embodiments, a strength of the centrifugal force depends on the velocity of the rotation **255** and rotating the substrate **220** faster generates a stronger centrifugal force that removes more developed material **206** or **212** from the top surface of the substrate **220** and generates less number of defects after the subsequent etching operation in the etched pattern **230**. In some embodiments, the etched pattern **230** is used for generating connection lines and the defects **232** and **234** are line collapses, e.g., pattern collapses.

(40) As described with respect to FIG. **1**B, after the water rinsing operation S**156**, the spin dry operation S**158** is applied to the substrate **220**. During the spin dry operation S**158** the substrate is rotated by the rotation **255** while no water is applied to the top surface or back side of the substrate **220**. Thus, during the spin dry operation S**158**, there is no barrier force in the direction **216** that blocks the developed material **206** from leaving the edge region **226**. Thus, during the spin dry operation S**158**, increasing the angular velocity of the rotation **255** may increase the centrifugal force caused by the rotation **255** of the substrate **220** may remove more developed material **206** or **212** from the top surface of the substrate **220**. FIG. **7**E shows a graph **750** of the number of line collapses, e.g., defects **232** or **234**, in the etched test pattern on a wafer, on a coordinate **714**, vs. the angular velocity of the rotation **255**, on a coordinate **712**. In some embodiments, the coordinate **712** shows the number of rotations per minute. As shown in FIG. **7**E, when the angular velocity of the rotation **255** changes from 1500 rotations per minute (rpm) to 3500 rpm, the number of line collapses, e.g., defects **232** and **234** in the etched pattern on the wafer changes from 32 to zero. In some embodiments, the spin dry operation S**158** is performed for between about 0.1 seconds and 5 seconds. In some embodiments the spin dry operation S**158** is performed for between about 5 seconds and 15 seconds. In some embodiments, the rotation **255** of the substrate **220** is set between about 100 cycles per minute to about 1500 cycles per minute.
(41) FIGS. **8**A, **8**B, **8**C, **8**D, **8**E, **8**F, and **8**G show a system for generating a resist pattern and/or an etched pattern on the substrate, an exhaust control system of the system for generating the resist pattern and/or the etched pattern on the substrate, and developed material residues and water residues with and without using the exhaust control system in accordance with some embodiments of the present disclosure. FIG. **8**A shows a photoresist development system **800** for developing the photoresist layer of the substrate **220** and rising the substrate **220** after the photoresist development. The photoresist development system **800** shows a gas blower system **810** and a linear nozzle **811** coupled to an enclosure **855** of the photoresist development system **800**. The linear nozzle **811**, through the gas blower system **810** and pipe **804**, is coupled to a gas source **816**. The gas source **816** is coupled to a developer controller **822** that generates and controls the flow of fan shape stream of gas **833**. The fan shape stream of gas **833** enters the enclosure **855** of the photoresist development system **800**. In some embodiments, at least a portion of the fan shape stream of gas **833** enters a development chamber **805** inside the enclosure **855** through an opening **832** of the development chamber **805**. The development chamber **805** shows a stage **825**, e.g., a rotatable wafer stage, and a wafer **830**, e.g., a photoresist-coated wafer, loaded on the stage **825** and standing on stage pins **856**. In some embodiments, the stage **825** rotates around an axis **854** with the rotation **255**.
(42) The development chamber **805** is connected via at least a cup **852** below the stage **825** to a gas transfer chamber **809**. In some embodiments, the cup **852** is formed with the shape of a hollow ring, e.g., a hollow cylinder **857** connected to a bottom of the development chamber **805** with another cylinder **853** inside hollow cylinder **857**. The cylinder **853** may be concentric with the hollow cylinder **857** and may be hollow or solid. In some embodiments, the axis **854** is mounted on top of the cylinder **853**. The portion of the fan shape stream of gas **833**, a purge gas, that enters the development chamber **805** and blows over the wafer **830**, goes down from the edge of the wafer **830** along a length of the axis **854**, and flows through the cup **852** (having a width **851**, e.g., the width of the hollow ring) to the gas transfer chamber **809**. In some embodiments, the flow rate of the fan shape stream of gas **833** that is controlled by the developer controller **822**, which determines the flow rates of the streams of the gas that blow over the wafer **830** and the stream of gas **836** that flow through the cups **852** to the gas transfer chamber **809**. The photoresist development system **800** also shows an exhaust system **820** that connects the gas transfer chamber **809** to an exhaust gas port **850** of the photoresist development system **800**. A stream of gas **846** leaves the gas transfer chamber **809** through a first entrance port **865** of the exhaust system **820** and enters the exhaust system **820**.

(43) A remaining portion of the fan shape stream of gas **833** that does not enter the development chamber **805** passes through the enclosure **855** and via a second entrance port **835** of the exhaust system **820** enters as a stream of gas **866** to the exhaust system **820**. The combination of the stream of gas **866** and the stream of gas **846** generates an exhaust stream of gas **834** that leaves the photoresist development system **800** through the exhaust gas port **850**. In some embodiments, the linear nozzle **811** blows the purge gas over the wafer **830** to the gas transfer chamber **809** that eventually exits the exhaust gas port **850**. As described, blowing the fan shape stream of gas **833** may add to the centrifugal force caused by the rotation **255** of the substrate **220** and move more developed material residues **224** or **228** from the top surface of the substrate **220** and generates less number of defects after the subsequent etching operation in the etched pattern **230**. In some embodiments, after the apply developer solution operation S**154**, the developed material **212** that includes dissolved photoresist is spread on the surface of the substrate **220** and the DI water in the water rinsing operation S**156** removes the dissolved photoresist from the surface of the substrate **220**. As described above, the water rinsing operation S**156** may not sufficiently clean the surface of the substrate **220** and some developed material residues and water residues remain on the surface of the substrate **220**.
(44) In some embodiments, the flow rates of the stream of gas **846**, which is the flow rate of the stream of gas **836** that pass through the cup **852**, is the flow rate that passes over the wafer **830** and is the flow rate that goes through the development chamber **805** and exits the gas transfer chamber **809** via the first entrance port **865** of the exhaust system **820**. Thus, the flow rate of the stream of gas **846** is an exit flow rate of the development chamber **805**, e.g., a wafer flow rate, which is a flow rate passing over the wafer, in the development chamber **805**. In some embodiments, as described with respect to FIG. **5**, in the development chamber **805**, before the water rinsing operation S**156**, a development material, consistent with the development material **57**, is applied in an apply developer solution operation S**154** to the to surface of the substrate **220** through a nozzle (not shown), consistent with the nozzle **62**. In some embodiments, the developer controller **822** is coupled to the stage **825** for controlling a spin rate of the stage **825**. In some embodiments, the developer controller **822** includes the stage controller **170**.
(45) In some embodiments, the developer controller **822** controls the flow rate of the fan shape stream of gas **833** exiting the linear nozzle **811** through controlling a pressure of the gas exiting the linear nozzle **811**. As shown in FIG. **8**A, the developer controller **822** is also coupled to the exhaust system **820** and by controlling the exhaust system **820** controls the flow rate of the exhaust stream of gas **834** exiting the exhaust gas port **850**.
(46) In some embodiments, the developed materials **212** or **206** include development material **57** and dissolved photoresist, and are spread on the substrate **220**. A purge gas is projected by the fan shape stream of gas **833** to the surface of the substrate **220**. In some embodiments, the purge gas is one or more gases selected from the group consisting of clean dry air, nitrogen, argon, helium, neon, and carbon dioxide. In some embodiments, the purge gas has less than about 1 ppb impurities and less than about 1% relative humidity.
(47) FIGS. **8**B and **8**C show the exhaust system **820**. The exhaust system **820** of FIGS. **8**B and **8**C includes the first entrance port **865** and the second entrance port **835**. The stream of gas **846** enters the exhaust system **820** through the first entrance port **865** and the stream of gas **866** enters the exhaust system **820** through the second entrance port **835**. The exhaust system **820** of FIGS. **8**B and **8**C includes the exhaust gas port **850** where the exhaust stream of gas **834**, a combination of the streams of gas **866** and **846**, leaves the exhaust system **820** through the exhaust gas port **850**. As shown in FIG. **8**B, the exhaust system **820** includes a hinged shutter **845** (e.g., a damper) that rotates around a hinge **840** in an angular direction **849**. A shutter controller **815** is coupled to and controls the hinged shutter **845**. By rotating the hinged shutter **845**, an angle T of the shutter with the horizontal direction changes and openings of the first and second entrance ports **865** and **835** are modified and the streams of gas **846** and **866** are adjusted. In some embodiments, the hinged

shutter **845** is rotated, the angle T becomes zero, the second entrance port **835** becomes closed, and the stream of gas **866** becomes zero, e.g., does not exist, and all the fan shape stream of gas **833** passes through the development chamber **805**, the cup **852**, and the gas transfer chamber **809** to the exhaust system **820**.

(48) In some embodiments, as the angle T increases, the second entrance port **835** becomes more open and the first entrance port **865** becomes less open. Thus, the flow rate of the stream of gas **846** is reduced and the flow rate of the stream of gas **866** is increased. In some embodiments, when the second entrance port **835** is closed, the pressure and the flow rate of the stream of gas **836**, **846**, or **834** is determined by the width **851** of the cup **852** and the pressure and the flow rate of the fan shape stream of gas **833**. In some embodiments, when the second entrance port **835** is partially or completely open, the pressure and the flow rate of the stream of gas **836** is determined by the width **851** of the cup **852**, the pressure and the flow rate of the fan shape stream of gas **833**, an amount of opening of the second entrance port **835**, e.g., the angle T of the hinged shutter **845**.

(49) As shown in FIG. **8**C, in some embodiments, the exhaust system **820** includes a sliding shutter **844** that moves over ball bearings **842** in a horizontal direction **847**. The shutter controller **815** is coupled to and controls the sliding shutter **844**. By moving the sliding shutter **844**, an opening size of the second entrance port **835** is modified and the streams of gas **846** and **866** are adjusted. In some embodiments, the sliding shutter **844** is moved such that the second entrance port **835** becomes closed, and the stream of gas **866** becomes zero, e.g., does not exist, and all the fan shape stream of gas **833** passes through the development chamber **805**, the cup **852**, and the gas transfer chamber **809** to the exhaust system **820**. As shown in FIG. **8**A, the developer controller **822** is coupled to the shutter controller **815** of exhaust system **820** and controls the flow rate of the streams of gas **833**, **836**, **846**, and **834**. In some embodiments, the flow rate of the stream of gas **846** (the exit flow rate) is proportional to the flow rate of the fan shape stream of gas **833**. In some embodiments, the flow rate of the stream of the exhaust stream of gas **834** is proportional to the flow rate of the fan shape stream of gas **833**. In some embodiments, the flow rate of the stream of gas **866** is proportional to the flow rate of the fan shape stream of gas **833**. In some embodiments, when the hinged shutter **845** or the sliding shutter **844** closes the second entrance port **835**, the flow rate of the stream of gas **866** is reduced and the flow rate of the stream of gas **846** is increased.

(50) In some embodiments, the opening size of the second entrance port **835** is reduced, the flow rate of the stream of gas **866** is reduced and the flow rate of the streams of gas **846** is increased. In some embodiments, the opening size of the second entrance port **835** is increased, the flow rate of the stream of gas **866** is increased and the flow rate of the streams of gas **846** is reduced. In some embodiments, the sliding shutter **844** is moved such that the second entrance port **835** becomes closed and the stream of gas **866** becomes zero, e.g., does not exist, and all the fan shape stream of gas **833** passes through the development chamber **805**, the cup **852**, and the gas transfer chamber **809** to the exhaust system **820**.

(51) In some embodiments, as the sliding shutter **844** moves to the right (opposite to the direction **847**), the second entrance port **835** becomes more open. Thus, the flow rate of the stream of gas **846** is reduced and the flow rate of the stream of gas **866** is increased and vice versa. In some embodiments, when the second entrance port **835** is closed, the pressure and the flow rate of the stream of gas **836**, **846**, or **834** is determined by the width **851** of the cup **852** and the pressure and the flow rate of the fan shape stream of gas **833**. In some embodiments, when the second entrance port **835** is partially or completely open, the pressure and the flow rate of the stream of gas **836** is determined by the width **851** of the cup **852**, the pressure and the flow rate of the fan shape stream of gas **833**, an amount of opening of the second entrance port **835**, e.g., where the sliding shutter **844** is placed.

(52) In some embodiments, a flow rate of the fan shape stream of gas **833**, e.g., the purge gas, during the applying the purge gas ranges from about 50 cc/s to about 2000 cc/s. In some embodiments, a flow rate of the purge gas during the applying the purge gas ranges from about 100

cc/s to about 1000 cc/s. In other embodiments, a flow rate of the purge gas during the applying a purge gas ranges from about 150 cc/s to about 500 cc/s. In some embodiments, the purge gas is applied to the substrate during the spin dry operation S**158** for about 10 seconds to about 20 minutes. In some embodiments, the purge gas is applied to the substrate for about 30 seconds to about 10 minutes. In some embodiments, the purge gas is applied to the substrate for about 1 minute to about 5 minutes. In some embodiments, the purge gas flow rate is varied (e.g., decreasing) during the applying the purge gas. For example, in some embodiments, a purge gas flow rate of about 200 cc/s is applied to the wafer for about 1 minute and then a purge gas flow rate of about 100 cc/s is applied for about 10 minutes. In another embodiment, a purge gas flow rate of about 1000 cc/s is applied for about 1 minute and then a purge gas flow rate of about 100 cc/s is applied for about 1 minute. In another embodiment, a purge gas flow rate of about 200 cc/s is applied for about 5 minutes and then a purge gas flow rate of about 100 cc/s is applied for about 5 minutes. At purge gas flow rates and purge gas flow times below the disclosed ranges, there may be insufficient removal of the residues. At purge gas flow rates and purge gas flow times greater than the disclosed ranges, there may be increased manufacturing costs with no significant improvement in the defect rate or device performance.

(53) In some embodiments, the pre-wet operation S**152**, is performed under a medium pressure gas flow setting of the development chamber **805** such that the developer controller **822** sets the exit flow rate of the stream of gas **846** of the development chamber **805** and the gas transfer chamber **809** to a medium flow rate of between about 150 cc/s and about 250 cc/s, e.g., about 200 cc/s, and the gas pressure of the development chamber **805** and the gas transfer chamber **809** is set to a medium pressure of between about 45 kilo Pascal (kPa) and about 65 kPa, e.g., about 55 kPa. Also, in some embodiments, the apply developer solution operation S**154** and the water rinse operation S**156**, are performed under the medium pressure gas flow setting in the development chamber **805** of the photoresist development system **800**. In some embodiments, the spin dry operation S**158**, is performed under a high pressure gas flow setting such that the developer controller **822** sets the exit flow rate of the stream of gas **846** of the development chamber **805** and the gas transfer chamber **809** to a high flow rate of between about 800 cc/s and about 1200 cc/s, e.g., about 1000 cc/s. In addition, the development chamber **805** and the gas transfer chamber **809** may have a high pressure, e.g., a pressure of the development chamber **805** and the gas transfer chamber **809** is set to a high pressure between about 120 kPa and about 160 kPa, e.g., about 140 kPa. In some embodiments, before loading the wafer **830** to the stage **825**, the development chamber **805** is set to a low pressure gas flow setting such that the developer controller **822** sets the exit flow rate of the stream of gas **846** of the development chamber **805** and the gas transfer chamber **809** to a low flow rate of between about 15 cc/s and about 25 cc/s, e.g., about 20 cc/s, and the development chamber **805** and the gas transfer chamber **809** are set to a low pressure of between about 4 kPa and about 6 kPa, e.g., about 5 kPa. In some embodiments, the developed material residues and water residues are drastically reduced when the spin dry operation S**158** is performed under the high pressure gas flow setting. In some embodiments, during the spin drying, the developer controller **822** modifies the exit flow rate of the stream of gas **846** of the development chamber **805** and the gas transfer chamber **809** between about 800 cc/s and about 1200 cc/s. In some embodiments, the developer controller **822** modifies the pressure of the development chamber **805** between about 120 kPa and about 160 kPa. In some embodiments, the developer controller **822** periodically alternates, e.g., between every 10 seconds to 2 minutes, the exit flow rate of the stream of gas **846** of the development chamber **805** and the gas transfer chamber **809** between about 800 cc/s and about 1200 cc/s. In some embodiments, the developer controller **822** periodically alternates, between every 10 seconds to 2 minutes, the pressure of the development chamber **805** between about 120 kPa and about 160 kPa.

(54) In some embodiments, when the development chamber **805** (and the gas transfer chamber **809**) is under the high pressure gas flow setting, the hinged shutter **845** is at zero degrees and the second

entrance port **835** of the exhaust system **820** is closed. In some embodiments, when the development chamber **805** (and the gas transfer chamber **809**) is under the low pressure gas flow setting, the hinged shutter **845** is at 90 degrees and the second entrance port **835** of the exhaust system **820** is completely open. In some embodiments, when the development chamber **805** (and the gas transfer chamber **809**) is under the medium pressure gas flow setting, the angle T of the hinged shutter **845** is at about 25-35 degrees (e.g., about 30 degrees) and the second entrance port **835** of the exhaust system **820** is partially open.

(55) In some embodiments, when the development chamber **805** (and the gas transfer chamber **809**) is under the high pressure gas flow setting, the sliding shutter **844** is completely moved in the direction **847** and the second entrance port **835** of the exhaust system **820** is closed. In some embodiments, when the development chamber **805** (and the gas transfer chamber **809**) is under the low pressure gas flow setting, the sliding shutter **844** is completely moved opposite to the direction **847** and the second entrance port **835** of the exhaust system **820** is completely open. In some embodiments, when the development chamber **805** (and the gas transfer chamber **809**) is under the medium pressure gas flow setting, the sliding shutter **844** is at about half way between completely open and completely closed and the second entrance port **835** of the exhaust system **820** is partially open. In some embodiments, under the medium pressure gas flow setting, the second entrance port **835** is between about 40 percent and 60 percent open.

(56) FIG. **8**D shows a map of developed material residues on the surface of the substrate **220** when the back side of the substrate **220** is rinsed in FIG. **7**A and the spin dry operation S**158** is performed under the medium pressure gas flow setting. FIG. **8**F shows a map of developed material residues on the surface of the substrate **220** for the same operation in FIG. **8**D except that the spin dry operation S**158** is performed under the high pressure gas flow setting. FIG. **8**E shows a map of developed material residues and water residues on the surface of the substrate **220** when the spin dry operation S**158** is performed under the medium pressure gas flow setting. FIG. **8**G shows a map of developed material residues and water residues on the surface of the substrate **220** for the same operation in FIG. **8**E except the spin dry operation S**158** is performed under the high pressure gas flow setting. As shown, performing the spin dry operation S**158** under the high pressure gas flow setting significantly reduces the developed material residues. In some embodiments when the angular velocity of the stage **825** is increased during the spin dry operation S**158**, the centrifugal force caused by the rotation **255** is increased and the developed material residues and water residues are decreased.

(57) In some embodiments, after the spin dry operation S**158**, the substrate is etched in etching operation S**114**. In etching operation S**114**, the remaining resist material, the resist pattern, is used as a mask and the exposed regions of the substrate is etched to produce an etched pattern on the substrate. In some embodiments, the pattern of openings **55***a*, **55***b* in the photoresist layer **15** (see FIGS. **6**A and **6**B) are extended into the semiconductor substrate **10** to create a pattern of openings **55***a*′, **55***b*′ in the semiconductor substrate **10**, thereby transferring the pattern in the photoresist layer **15** into the semiconductor substrate **10**, as shown in FIGS. **9**A and **9**B. The pattern is extended into the substrate by etching, using one or more suitable etchants. The remaining photoresist of the region **50**, **52** is at least partially removed during the etching operation in some embodiments. In other embodiments, the remaining photoresist of the region **50**, **52** is removed after etching the semiconductor substrate **10** by using a suitable photoresist stripper solvent or by a photoresist ashing operation.

(58) In some embodiments, the resist layer on the surface of the substrate is inspected after the development operation S**112** in an after-development inspection (ADI) operation S**122** and the developed material residues on the surface of the wafer are mapped. In some embodiments, the etched layer on the surface of the substrate is inspected after the etching operation S**114** in an after etching inspection (AEI) operation S**123** and the presence of any etching defects and developed material residues on the surface of the wafer are mapped.

(59) In some embodiments, the semiconductor substrate **10** includes a single crystalline semiconductor layer on at least its surface portion. The semiconductor substrate **10** may include a single crystalline semiconductor material such as, but not limited to Si, Ge, SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb and InP. In some embodiments, the semiconductor substrate **10** is a silicon layer of an SOI (silicon-on insulator) substrate. In certain embodiments, the semiconductor substrate **10** is made of crystalline Si.
(60) The semiconductor substrate **10** may include in its surface region, one or more buffer layers (not shown). The buffer layers can serve to gradually change the lattice constant from that of the substrate to that of subsequently formed source/drain regions. The buffer layers may be formed from epitaxially grown single crystalline semiconductor materials such as, but not limited to Si, Ge, GeSn, SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, GaN, and InP. In an embodiment, the silicon germanium (SiGe) buffer layer is epitaxially grown on the semiconductor substrate **10**. The germanium concentration of the SiGe buffer layers may increase from 30 atomic % for the bottom-most buffer layer to 70 atomic % for the top-most buffer layer.
(61) In some embodiments, the semiconductor substrate **10** includes one or more layers of at least one metal, metal alloy, and metal nitride/sulfide/oxide/silicide having the formula MX.sub.a, where M is a metal and X is N, S, Se, O, Si, and a is from about 0.4 to about 2.5. In some embodiments, the semiconductor substrate **10** includes titanium, aluminum, cobalt, ruthenium, titanium nitride, tungsten nitride, tantalum nitride, and combinations thereof.
(62) In some embodiments, the semiconductor substrate **10** includes a dielectric material having at least a silicon or metal oxide or nitride of the formula MX.sub.b, where M is a metal or Si, X is N or O, and b ranges from about 0.4 to about 2.5. In some embodiments, the semiconductor substrate **10** includes silicon dioxide, silicon nitride, aluminum oxide, hafnium oxide, lanthanum oxide, and combinations thereof.
(63) The photoresist layer **15** is a photosensitive layer that is patterned by exposure to actinic radiation. Typically, the chemical properties of the photoresist regions struck by incident radiation change in a manner that depends on the type of photoresist used. Photoresist layers **15** are either positive tone resists or negative tone resists. A positive tone resist refers to a photoresist material that when exposed to radiation, such as UV light, becomes soluble in a developer, while the region of the photoresist that is non-exposed (or exposed less) is insoluble in the developer. A negative tone resist, on the other hand, refers to a photoresist material that when exposed to radiation becomes insoluble in the developer, while the region of the photoresist that is non-exposed (or exposed less) is soluble in the developer. The region of a negative resist that becomes insoluble upon exposure to radiation may become insoluble due to a cross-linking reaction caused by the exposure to radiation.
(64) Whether a resist is a positive tone or negative tone may determine the type of developer used to develop the resist. For example, some positive tone photoresists provide a positive pattern, (i.e. —the exposed regions are removed by the developer), when the developer is an aqueous-based developer, such as a tetramethylammonium hydroxide (TMAH) solution. On the other hand, the same photoresist provides a negative pattern (i.e. —the unexposed regions are removed by the developer) when the developer is an organic solvent, such as n-butyl acetate (nBA). Further, in some negative tone photoresists developed with the TMAH solution, the unexposed regions of the photoresist are removed by the TMAH, and the exposed regions of the photoresist, that undergo cross-linking upon exposure to actinic radiation, remain on the substrate after development.
(65) In some embodiments, the photoresist layer **15** includes a high sensitivity photoresist composition. In some embodiments, the high sensitivity photoresist composition is highly sensitive to extreme ultraviolet (EUV) radiation. In some embodiments, the photoresist composition includes a polymer, a photoactive compound (PAC), and a sensitizer. In some embodiments, the photoresist includes metal nanoparticles.
(66) In some embodiments, the photoresist layer **15** is a tri-layer resist. A tri-layer resist includes a

bottom layer, a middle layer, and an upper layer. In some embodiments, the bottom layer is a planarizing layer or a bottom anti-reflective coating (BARC) layer. In some embodiments, the bottom layer is formed of a carbon backbone polymer. In some embodiments, the middle layer is a made of a silicon-containing material. In some embodiments, the middle layer is an anti-reflective layer. The upper layer is a photosensitive layer that is patterned like the photoresist layers described herein.

(67) FIGS. **10**A and **10**B show an inspection system **600** for inspecting residue or defects on a surface of the substrate and a system that controls gas purging of the substrate in accordance with some embodiments of the present disclosure. In some embodiments, the residues are the developed material residues and/or the water residues on the developed resist pattern of the substrate **220** and the defects are the defects of the etched pattern **230** on the substrate **220**. FIG. **10**A shows a scanning-imaging device **635** and a lens **634** that generates a uniform light beam **617** for imaging the top surface of the substrate **220** to generate an image of the top surface of substrate **220**. In addition, FIG. **10**A shows the scanning-imaging device **635** generates a focusing beam **619** for scanning a top surface of the substrate **220** to generate an scanned image of the top surface of the substrate **220**. The scanning-imaging device **635** is coupled to an analyzer module **630** that receives the images captured by the scanning-imaging device **635**. The analyzer module **630** includes an image processing unit **633** for processing the received images. In FIG. **10**A, the substrate **220** is disposed on the stage **520**. The stage **520** is coupled and controlled by a stage controller **665**. The scanning-imaging device **635** is also couple to the stage controller **665** and receives a location of the wafer being scanned and/or imaged. The scanning-imaging device **635** captures one or more images of the surface of the substrate **220** at different locations of the substrate **220** and sends the images and the locations to the analyzer module **630** or the image processing unit **633**. The analyzer module **630** or the image processing unit **633** of the analyzer module **630** determines the number of residues or defects and locations of the residue or defects on the substrate **220**, e.g., determines a map of the residues or defects.

(68) FIG. **10**B shows a purge system **650** that includes the inspection system **600** coupled to the photoresist development system **800**, e.g., to the developer controller **822** of the photoresist development system **800**. In some embodiments, if the inspection system **600** determines that the number of residues, or a density of the residues, e.g., residues per square millimeter of the surface of the substrate **220**, is above a threshold value, the inspection system **600** sends information **702** to the developer controller **822** and the developer controller **822** increases the duration of the next spin dry operation S**158**, increases the flow rate of the purge gas, or increases the pressure of the purge gas during the next spin dry operation S**158**. In some embodiments, if the number of residues, or a density of the residues, on the surface of the substrate **220**, is above the threshold value, the inspection system **600** sends the information to the developer controller **822** and the wafer is sent back to the photoresist development system **800** for further water rinse operation S**156** and spin dry operation S**158**. The developer controller **822** increases the duration of the spin dry operation S**158**, increases the flow rate of the purge gas, or increases the pressure of the purge gas during the spin dry operation S**158**. In some embodiments, the threshold value of the residue is greater than 1 residue/mm.sup.2. In some embodiments, the threshold value of the residue is greater than 0.5 residues/mm.sup.2. In other embodiments, the threshold value of the residue is greater than 0.1 residues/mm.sup.2.

(69) In some embodiments, if the number of defects, or a density of the defects, e.g., defects per square millimeter of the etched pattern on the surface of the substrate **220**, is above a threshold value, the inspection system **600** sends the information to the developer controller **822** and the developer controller **822** increases the duration of the next spin dry operation S**158**, increases the flow rate of the purge gas, or increases the pressure of the purge gas during the next spin dry operation S**158**.

(70) FIG. **11** shows a control system **700** for generating a resist pattern and/or an etched pattern on

the substrate in accordance with some embodiments of the present disclosure. The control system **700** includes the analyzer module **630** and a main controller **740** coupled to each other. In some embodiments, the main controller **740** is programmed to: control a spin rate of the stages **160**, **520**, **825**; dispensing of liquid or gas from any of the nozzles **62**, **208**, **811**; translational motion of any of the nozzles **62**, **208**, **811**; and a tilt angle of any of the nozzles **62**, **208**, **811** relative to surfaces of the substrate. As shown in FIG. **11**, the main controller **740** is programmed to control one or more of the analyzer module **630**, and the image processing unit **633** and the scanning-imaging device **635** through the analyzer module **630**. In addition, the main controller **740** is programmed to control the stage controllers **170**, **665**; the various nozzles **62**, **208**, **811**; the developer controller **822**; the vacuum pressure controller **406**; and the exposure controller **430**. In some embodiments, the main controller **740** includes the developer controller **822** and in some embodiments, the developer controller **822** controls the scanning-imaging device **635** and the analyzer module **630**.

(71) FIG. **12** shows a flow diagram of a process **1200** for generating a resist pattern and/or an etched pattern on a substrate in accordance with some embodiments of the present disclosure. In some embodiments, a photoresist layer **15** on a wafer, e.g., the semiconductor substrate **10**, is formed and a photoresist coated wafer is produced in operation S**1210**. The photoresist layer **15** is selectively exposed to actinic radiation to form a latent pattern in the photoresist layer **15** in operation S**1220**. The latent pattern is developed by applying a developer, a development material **57**, to the selectively exposed photoresist layer **15** as shown in FIG. **4**A in operation S**1230**. The photoresist layer **15** is rinsed to form a patterned photoresist layer exposing a portion of the wafer as shown in FIGS. **6**A and **6**B in operation S**1240**. The patterned photo resist layer, e.g., the patterned photoresist layer of FIGS. **6**A and **6**B is spin dried under the high pressure gas flow setting to remove developed material residues from the patterned photoresist layer in operation S**1250**.

(72) The resist layer is developed with a photoresist developer (the development material **57**) after selectively exposing the resist layer to actinic radiation in operation S**1230**. The developed photoresist layer **15** is cleaned with a cleaning liquid, such as deionized water, in operation S**1240**. Then, in operation S**1250**, a gas purge is applied to the semiconductor substrate **10** to remove residual developed material and cleaning liquid. In some embodiments, the semiconductor substrate **10** is subsequently inspected to verify the sufficiency of the gas purge operation.

(73) FIGS. **13**A and **13**B illustrate an apparatus for controlling the method for generating a resist pattern and/or an etched pattern in accordance with some embodiments of the present disclosure. In some embodiments, the computer system **900** is used for performing the functions of the modules of FIG. **11** that include the main controller **740**, the analyzer module **630**, the stage controller **665** or **170**, the vacuum pressure controller **406**, and the image processing unit **633** that is included in the analyzer module **630**. In some embodiments, the computer system **900** is used to execute the process **100** of FIG. **1** and process **1200** of FIG. **12**.

(74) FIG. **13**A is a schematic view of a computer system that performs the functions of a controller for controlling the method for generating a resist pattern and/or an etched pattern, including the gas purge of a substrate. All of or a part of the processes, method and/or operations of the foregoing embodiments can be realized using computer hardware and computer programs executed thereon. In FIG. **13**A, a computer system **900** is provided with a computer **1001** including an optical disk read only memory (e.g., CD-ROM or DVD-ROM) drive **1005** and a magnetic disk drive **1006**, a keyboard **1002**, a mouse **1003**, and a monitor **1004**.

(75) FIG. **13**B is a diagram showing an internal configuration of the computer system **900**. In FIG. **13**B, the computer **1001** is provided with, in addition to the optical disk drive **1005** and the magnetic disk drive **1006**, one or more processors, such as a micro processing unit (MPU) **1011**, a ROM **1012** in which a program such as a boot up program is stored, a random access memory (RAM) **1013** that is connected to the MPU **1011** and in which a command of an application program is temporarily stored and a temporary storage area is provided, a hard disk **1014** in which

an application program, a system program, and data are stored, and a bus **1015** that connects the MPU **1011**, the ROM **1012**, and the like. Note that the computer **1001** may include a network card (not shown) for providing a connection to a LAN.

(76) The program for causing the computer system **900** to execute the functions for generating a resist pattern and/or an etched pattern in the foregoing embodiments may be stored in an optical disk **1021** or a magnetic disk **1022**, which are inserted into the optical disk drive **1005** or the magnetic disk drive **1006**, and transmitted to the hard disk **1014**. Alternatively, the program may be transmitted via a network (not shown) to the computer **1001** and stored in the hard disk **1014**. At the time of execution, the program is loaded into the RAM **1013**. The program may be loaded from the optical disk **1021** or the magnetic disk **1022**, or directly from a network. The program does not necessarily have to include, for example, an operating system (OS) or a third party program to cause the computer **1001** to execute the functions of the control system for generating a resist pattern and/or an etched pattern in the foregoing embodiments. The program may only include a command portion to call an appropriate function (module) in a controlled mode and obtain desired results.

(77) FIG. **14** shows a process stage of a sequential operation according to an embodiment of the disclosure. As shown, in some embodiments, the substrate **68** is an in-process substrate **68** and a layer to be patterned **60** is disposed over a bare semiconductor substrate **10** prior to forming the photoresist layer. In some embodiments, the layer to be patterned **60** is a metallization layer or a dielectric layer, such as a passivation layer, disposed over a metallization layer. In embodiments where the layer to be patterned **60** is a metallization layer, the layer to be patterned **60** is formed of a conductive material using metallization processes, and metal deposition techniques, including chemical vapor deposition, atomic layer deposition, and physical vapor deposition (sputtering). Likewise, if the layer to be patterned **60** is a dielectric layer, the layer to be patterned **60** is formed by dielectric layer formation techniques, including thermal oxidation, chemical vapor deposition, atomic layer deposition, and physical vapor deposition.

(78) FIGS. **15**A and **15**B show a process stage of a sequential operation according to an embodiment of the disclosure. The photoresist layer **15** is subsequently selectively exposed or patternwise exposed to actinic radiation **45** to form exposed regions **50** and unexposed regions **52** in the photoresist layer **15**, as shown in FIGS. **15**A and **15**B, as described herein in relation to FIGS. **4**A and **4**B. As shown in FIG. **15**B, the EUV patterned beam **31**, reflected from the photomask **205***c*, is directed to the photoresist layer **15**.

(79) FIG. **16** shows a process stage of a sequential operation according to an embodiment of the disclosure. As shown in FIG. **16**, the selectively exposed or patternwise exposed photoresist layer **15** is developed by dispensing development material **57** from a nozzle **62** to form a pattern of photoresist openings to produce the pattern of openings **55***a*, **55***b*, in the photoresist layer over the in-process substrate **68**, as shown in FIGS. **17**A and **17**B, as described herein in relation to FIGS. **5**, **6**A, and **6**B.

(80) FIGS. **17**A and **17**B show a process stage of a sequential operation according to an embodiment of the disclosure. FIG. **17**A illustrates the development of a positive tone photoresist pattern, and FIG. **17**B illustrates the development of a negative tone photoresist pattern. The resist patterns are generated using the photoresist development system **800** of FIG. **8**A.

(81) FIGS. **18**A and **18**B show a process stage of a sequential operation according to an embodiment of the disclosure. Then, as shown in FIGS. **18**A and **18**B, the pattern of openings **55***a*, **55***b* (see FIGS. **17**A and **17**B) in the photoresist layer **15** is transferred to the layer to be patterned **60** using an etching operation and the photoresist layer is removed, as explained with reference to FIGS. **9**A and **9**B to form the pattern of openings **55***a*″, **55***b*″ in the layer to be patterned **60**.

(82) The techniques to apply a purge gas at high pressure or flow rate described herein are not limited to removing developed material residues from semiconductor substrates. In some embodiments, the techniques disclosed herein are used to remove excess amounts and residues of

other coatings. For example, in some embodiments, the techniques are used to remove excess or residues of photoresist coatings; polymeric insulating layers, including polyimide layers; bottom anti-reflective coating (BARC) layers; top anti-reflective coating (TARC) layers; and spin-on-glass (SOG) layers. The gas purge techniques can be used to blow away any chemicals from the surface of the substrate.

(83) Other embodiments include other operations before, during, or after the operations described above. In some embodiments, the disclosed methods include forming semiconductor devices, including fin field effect transistor (FinFET) structures. In some embodiments, a plurality of active fins are formed on the semiconductor substrate. Such embodiments, further include etching the substrate through the openings of a patterned hard mask to form trenches in the substrate; filling the trenches with a dielectric material; performing a chemical mechanical polishing (CMP) process to form shallow trench isolation (STI) features; and epitaxy growing or recessing the STI features to form fin-like active regions. In some embodiments, one or more gate electrodes are formed on the substrate. Some embodiments include forming gate spacers, doped source/drain regions, contacts for gate/source/drain features, etc. In other embodiments, a target pattern is formed as metal lines in a multilayer interconnection structure. For example, the metal lines may be formed in an inter-layer dielectric (ILD) layer of the substrate, which has been etched to form a plurality of trenches. The trenches may be filled with a conductive material, such as a metal; and the conductive material may be polished using a process such as chemical mechanical planarization (CMP) to expose the patterned ILD layer, thereby forming the metal lines in the ILD layer. The above are non-limiting examples of devices/structures that can be made and/or improved using the method described herein.

(84) In some embodiments, active components such diodes, field-effect transistors (FETs), metal-oxide semiconductor field effect transistors (MOSFET), complementary metal-oxide semiconductor (CMOS) transistors, bipolar transistors, high voltage transistors, high frequency transistors, FinFETs, other three-dimensional (3D) FETs, other memory cells, and combinations thereof are formed, according to embodiments of the disclosure.

(85) As described in the foregoing embodiments, at the end of the development process the substrate undergoes the spin dry operation S**158** such that developed material residues and/or DI wafer residues are removed from the surface of the wafer. Removal of the residues cause the layout pattern to form, without error or defect, on a resist layer on the surface of the wafer and, thus, when the resist layer is used for etching, the defects in etched pattern is reduced. In some embodiments, the amount of residue is significantly reduced, such that there are only several residue particles remaining on a wafer. In some embodiments, the amount of residue particles on the wafer after the spin dry operation S**158** is less than 0.1 particles/mm.sup.2. In some embodiments, no residue particles are detected on a wafer after the spin dry operation S**158**. In some embodiments, the methods described above removes the DI water that has splashed from the backside, e.g., removes the effects of back splashing. In some embodiments, the methods and systems described above, prevents line collapse and hole blinding, enlarges the processing window, reduces the number of defects, improves critical dimension uniformity, and causes yield improvement.

(86) According to some embodiments of the present disclosure, a method for manufacturing a semiconductor device includes forming a photoresist layer including a photoresist composition over a wafer to produce a photoresist-coated wafer and selectively exposing the photoresist layer to actinic radiation to form a latent pattern in the photoresist layer. The method also includes developing the latent pattern by applying a developer to the selectively exposed photoresist layer under a first pressure gas flow setting in a development chamber and rinsing the photoresist layer to form a patterned photoresist layer exposing a portion of the wafer under the first pressure gas flow setting in the development chamber. The method also includes spin drying the patterned photoresist layer under a second pressure gas flow setting in the development chamber. A pressure of the development chamber under the second pressure gas flow setting is greater than the pressure of the

development chamber under the first pressure gas flow setting. In an embodiment, the method further includes flowing a purge gas from a nozzle into an enclosure of a development system that includes the development chamber inside the enclosure. The purge gas flows over the wafer during the developing, the rinsing, and the spin drying. A gas flow rate of the purge gas over the wafer in the development chamber under the second pressure gas flow setting is greater than the gas flow rate under the first pressure gas flow setting. In an embodiment, the method further includes opening a first entrance port of a shutter system coupled to an exit port of the development chamber to modify the first pressure gas flow setting to the second pressure gas flow setting. In an embodiment, the method further includes after the spin drying, determining an amount of developed material residue on the wafer, and when the amount of developed material residue exceeds a threshold amount, changing a parameter of the second pressure gas flow setting for a next spin drying. In an embodiment, one or more parameters of the second pressure gas flow setting include at least one of a flow rate setting in the development chamber or a pressure setting of the development chamber. In an embodiment, the purge gas is one or more gases selected from the group consisting of clean dry air, nitrogen, argon, helium, neon, and carbon dioxide. In an embodiment, the wafer is rotated while the purge gas is applied during the spin drying at a speed between about 100 rpm and about 2000 rpm. In an embodiment, the first pressure gas flow rate of a purge gas during the spin drying in the development chamber is between about 800 cc/s and 1200 cc/s and the pressure of the development chamber is between about 120 kPa and 140 kPa, and the second pressure gas flow setting of the flow rate of the purge gas during the spin drying in the development chamber is between about 250 cc/s and 150 cc/s and the pressure of the development chamber is between about 45 kPa and 65 kPa. In an embodiment, the shutter system includes a second entrance port coupled to the enclosure of the development system outside the development chamber. A remaining portion of the purge gas flows through the second entrance port to the shutter system and the method further includes reducing an opening size of the second entrance port to increase the flow rate of the purge gas in the development chamber.

(87) According to some embodiments of the present disclosure, a method for manufacturing a semiconductor device includes forming a photoresist layer over an in-process substrate. The in-process substrate includes a plurality of devices and conductive power and signal routing interconnections for the plurality of devices. The method also includes patternwise exposing the photoresist layer to actinic radiation to form a latent pattern in the photoresist layer and applying a developer solution to the latent pattern to form a pattern in the photoresist layer under a first pressure gas flow setting in a development chamber. The method further includes applying deionized water to the pattern after applying the developer solution to form a patterned photoresist layer under a first pressure gas flow setting in a development chamber and spin drying the patterned photoresist layer under a second pressure gas flow setting in the development chamber to reduces developed material residues and water residues from the patterned photoresist layer on the in-process substrate. A pressure of the development chamber under the second pressure gas flow setting is greater than the pressure of the development chamber under the first pressure gas flow setting. In an embodiment, the method further includes flowing a gas over the in-process substrate during the applying the developer, the applying the deionized water, and the spin drying. A gas flow rate of the gas in the development chamber under the second pressure gas flow setting is greater than the gas flow rate of the gas under the first pressure gas flow setting. In an embodiment, the applying the developer, the applying the deionized water, and the spin drying are performed in a development chamber, parameters of the second pressure gas flow setting includes setting a flow rate of the gas in the development chamber and setting a pressure of the development chamber, and during the spin drying, a gas flow rate in the development chamber is between about 800 cc/s and 1200 cc/s and a pressure in the development chamber is between about 120 kPa and 160 kPa. In an embodiment, the method further includes periodically alternating the pressure of the development chamber between about 120 kPa and about 160 kPa and periodically alternating the gas flow rate

between 800 cc/s and about 1200 cc/s during the spin drying. In an embodiment, the spin drying is performed for about 10 seconds to about 20 minutes.

(88) According to some embodiments of the present disclosure, a photoresist development system of a system for semiconductor device manufacturing includes a development chamber that includes a rotatable wafer stage to support a photoresist-coated wafer. The system also includes a nozzle disposed above the development chamber to apply a first stream of gas over the rotatable wafer stage in the development chamber, an exhaust gas port, and an exhaust system that includes a movable shutter coupled between the development chamber and the exhaust gas port. The system further includes a controller coupled through a gas source to the nozzle, coupled to the rotatable wafer stage, and coupled to the exhaust system, the controller performs: control a spin rate of the rotatable wafer stage, control a flow rate of a second stream of gas exiting from the nozzle, and control the movable shutter of the exhaust system such that by adjusting a location of the movable shutter, the controller adjusts a flow rate of the first stream of gas entering the development chamber and to adjust a pressure of the development chamber. In an embodiment, the photoresist development system includes a body and an enclosure surrounded by the body, the development chamber is located inside the enclosure and occupies a first portion of the enclosure, the exhaust system includes a first entrance port coupled to the development chamber and a second entrance port coupled to a second portion of the enclosure outside the development chamber, and by moving the movable shutter a ratio of a flow rate of the first stream of gas entering the development chamber to a flow rate of a third stream of gas entering the second portion of the enclosure is adjusted and the pressure of the development chamber is adjusted. In an embodiment, the movable shutter is a damper that rotates around a hinge and by rotating around the hinge, the damper opens and/or closes the second entrance port and adjusts the ratio of the flow rate of the first stream of gas to the flow rate of the third stream of gas and adjust the pressure of the development chamber. In an embodiment, the movable shutter is a sliding shutter that moves horizontally and by moving, the movable shutter opens and/or closes the second entrance port and adjust the ratio of the flow rate of the first stream of gas to the flow rate of the third stream of gas and adjusts the pressure of the development chamber. In an embodiment, the system further includes an inspection tool to inspect a surface of the photoresist-coated wafer. The inspection tool includes a wafer inspection support stage, a scanning-imaging device, and an analyzer module. The controller controls the wafer inspection support stage, the scanning/imaging device, and the analyzer module. In an embodiment, the system further includes adjusting the flow rate of the second stream of gas to adjust the flow rate of the first stream of gas entering the development chamber and to adjust the pressure of the development chamber.

(89) As described in the foregoing embodiments, at the end of the development process during spin drying the substrate is purged by high pressure gas such that developed material and/or DI wafer residues are further removed from the surface of the wafer. Removal of the residues cause the layout pattern to form, without error or defect, on a resist layer on the surface of the wafer and, thus, when the resist layer is used for etching, the defects in etched pattern is reduced and the critical dimension uniformity increases.

(90) The foregoing outlines features of several embodiments or examples so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments or examples introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for manufacturing a semiconductor device, comprising: forming a photoresist layer comprising a photoresist composition over a wafer to produce a photoresist-coated wafer; selectively exposing the photoresist layer to actinic radiation to form a latent pattern in the photoresist layer; developing the latent pattern by applying a developer to the selectively exposed photoresist layer under a first pressure gas flow setting in a development chamber; rinsing the photoresist layer to form a patterned photoresist layer exposing a portion of the wafer under the first pressure gas flow setting in the development chamber; and spin drying the patterned photoresist layer under a second pressure gas flow setting in the development chamber, wherein a pressure of the development chamber under the second pressure gas flow setting is greater than the pressure of the development chamber under the first pressure gas flow setting.
2. The method of claim 1, further comprising: flowing a purge gas from a nozzle into an enclosure of a development system that includes the development chamber inside the enclosure, wherein the purge gas flows over the wafer during the developing, the rinsing, and the spin drying, wherein a gas flow rate of the purge gas over the wafer in the development chamber under the second pressure gas flow setting is greater than the gas flow rate under the first pressure gas flow setting.
3. The method of claim 2, further comprising: opening a first entrance port of a shutter system coupled to an exit port of the development chamber to modify the first pressure gas flow setting to the second pressure gas flow setting.
4. The method of claim 3, further comprising: after the spin drying, determining an amount of developed material residue on the wafer, and wherein when the amount of developed material residue exceeds a threshold amount, changing a parameter of the second pressure gas flow setting for a next spin drying.
5. The method of claim 4, wherein one or more parameters of the second pressure gas flow setting comprise at least one of a flow rate setting in the development chamber or a pressure setting of the development chamber.
6. The method of claim 2, wherein the purge gas is one or more gases selected from the group consisting of clean dry air, nitrogen, argon, helium, neon, and carbon dioxide.
7. The method of claim 6, wherein the wafer is rotated while the purge gas is applied during the spin drying at a speed between about 100 rpm and about 2000 rpm.
8. The method of claim 5, wherein the first pressure gas flow setting comprises the gas flow rate of the purge gas during the spin drying in the development chamber is between about 800 cc/s and 1200 cc/s and the pressure of the development chamber is between about 120 kPa and 140 kPa, and wherein the second pressure gas flow setting comprises the gas flow rate of the purge gas during the spin drying in the development chamber is between about 250 cc/s and 150 cc/s and the pressure of the development chamber is between about 45 kPa and 65 kPa.
9. The method of claim 3, wherein the shutter system comprises a second entrance port coupled to the enclosure of the development system outside the development chamber, and wherein a remaining portion of the purge gas flows through the second entrance port to the shutter system, and the method further comprises reducing an opening size of the second entrance port to increase the gas flow rate of the purge gas in the development chamber.
10. A method for manufacturing a semiconductor device, comprising: forming a photoresist layer over an in-process substrate, wherein the in-process substrate comprises a plurality of devices and conductive power and signal routing interconnections for the plurality of devices; patternwise exposing the photoresist layer to actinic radiation to form a latent pattern in the photoresist layer; applying a developer solution to the latent pattern to form a pattern in the photoresist layer under a first pressure gas flow setting in a development chamber; applying deionized water to the pattern after applying the developer solution to form the patterned photoresist layer under the first pressure

gas flow setting in the development chamber; and spin drying the patterned photoresist layer under a second pressure gas flow setting in the development chamber to reduce developed material residues and water residues from the patterned photoresist layer on the in-process substrate, wherein a pressure of the development chamber under the second pressure gas flow setting is greater than the pressure of the development chamber under the first pressure gas flow setting.
11. The method according to claim 10, further comprising: flowing a gas over the in-process substrate during the applying the developer solution, the applying the deionized water, and the spin drying, wherein a gas flow rate of the gas in the development chamber under the second pressure gas flow setting is greater than the gas flow rate of the gas under the first pressure gas flow setting.
12. The method according to claim 10, wherein: the applying the developer solution, the applying the deionized water, and the spin drying are performed in the development chamber, parameters of the second pressure gas flow setting comprise setting a gas flow rate of a gas in the development chamber and setting a pressure of the development chamber, and during the spin drying, the gas flow rate in the development chamber is between about 800 cc/s and 1200 cc/s and a pressure in the development chamber is between about 120 kPa and 160 kPa.
13. The method of claim 12, further comprising: during the spin drying, periodically alternating the pressure of the development chamber between about 120 kPa and about 160 kPa and periodically alternating the gas flow rate between 800 cc/s and about 1200 cc/s.
14. The method according to claim 11, wherein the spin drying is performed for about 10 seconds to about 20 minutes.
15. A method for manufacturing a semiconductor device, comprising: forming a photoresist layer over a wafer; selectively exposing the photoresist layer to actinic radiation to form a latent pattern in the photoresist layer; developing the latent pattern under a first pressure gas flow setting in a development chamber; rinsing the photoresist layer to form a patterned photoresist layer exposing a portion of the wafer under the first pressure gas flow setting in the development chamber; and spin drying the patterned photoresist layer under a second pressure gas flow setting in the development chamber, wherein a pressure of the development chamber under the second pressure gas flow setting is greater than the pressure of the development chamber under the first pressure gas flow setting, and during the spin drying, periodically alternating the pressure of the development chamber under the second pressure gas flow setting between a first pressure and a second pressure, and a second pressure is greater than the first pressure.
16. The method of claim 15, further comprising: flowing a purge gas from a nozzle into an enclosure of a development system that includes the development chamber inside the enclosure, wherein the purge gas flows over the wafer during the developing, the rinsing, and the spin drying, wherein a gas flow rate of the purge gas over the wafer in the development chamber under the second pressure gas flow setting is greater than the gas flow rate under the first pressure gas flow setting.
17. The method of claim 16, further comprising: opening a first entrance port of a shutter system coupled to an exit port of the development chamber to modify the first pressure gas flow setting to the second pressure gas flow setting.
18. The method of claim 17, further comprising: after the spin drying, determining an amount of developed material residue on the wafer, and wherein when the amount of developed material residue exceeds a threshold amount, changing a parameter of the second pressure gas flow setting for a next spin drying.
19. The method of claim 18, wherein one or more parameters of the second pressure gas flow setting comprise at least one of a flow rate setting in the development chamber or a pressure setting of the development chamber.
20. The method of claim 16, wherein the purge gas is at least one gas selected from the group consisting of clean dry air, nitrogen, argon, helium, neon, and carbon dioxide.